

Matthew Broglio

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Technical Skills

- Software Tools: SOLIDWORKS, AutoCAD, Matterport, Bluebeam Revu, Newforma, MS Office
- Programming: C, Python, MATLAB

Education

Bachelor of Applied Science, Mechanical and Materials Engineering | Queen's University

September 2022 – April 2027

- Currently attending fourth year
- Deans Scholar (2023 – 2025) - Awarded for outstanding academic performance

Relevant Coursework

- Solid Mechanics 1, Solid Mechanics 2
- Thermodynamics 1, Thermodynamics 2
- Fluid Mechanics 1, Fluid Mechanics 2
- Engineering Design & Practice

Engineering Experience

Project Engineering Intern | TWD Technologies Inc.

Burlington, ON, May 2025 – August 2026

Suncor Tertiary Network VOC Project

- Collaborated with piping and mechanical engineering teams to support a multi-site Vapor Organic Compound (VOC) emissions reduction project across 41 petroleum bulk plants throughout Canada.
- Supported the engineering design and implementation of Vapor Balance Lines (VBLs) and Vapor Recovery Units (VRUs) by developing and updating piping layouts using Bluebeam Revu.
- Performed layout assessments of each site using Matterport 3D scans to determine optimal equipment locations, routing, and approximate piping lengths.
- Prepared and maintained Bills of Materials (BOMs) and updated Piping & Instrumentation Diagrams (P&IDs) to reflect the engineering design revisions throughout the project.

Trans-Northern Pipelines Inc. Obsolescence Project

- Supported a lifecycle assessment of 5000+ pipeline assets by reviewing Piping and Instrumentation Diagrams (P&IDs), single-line diagrams (SLDs), field inspections, and asset management records to determine equipment obsolescence status.
- Consolidated field-collected equipment data including asset ages, nameplate information, and technical documentation, into a centralized engineering database while resolving discrepancies with client engineering teams.
- Worked with equipment manufacturers and vendors to obtain equipment obsolescence status, replacement recommendations, lead times, and spare parts information.
- Developed and maintained the Bill of Materials, vendor databases, and project tracking tools throughout the duration of the project.

Thermal Team Member | Relectric Design Team, Queens University

Kingston, ON, September 2024 – April 2025

- Collaborated with four sub-teams to retrofit a 1997 Jeep TJ with an electrical vehicle system, to convert into a sustainably electric-powered vehicle.
- Conducted calculations and research to optimize the heat discharge and performance of the thermal system.
- Simulated heat management systems to improve overall efficiency in the system.

Vehicle Dynamics Team Member | Baja SAE Design Team, Queens University

Kingston, ON, September 2024 – April 2025

- Worked with a multi-disciplinary team that designs, fabricates, and races an off-road vehicle at international competitions.
- Designed and analyzed bent control arms for the front suspension in SOLIDWORKS, using FEA and 3D modeling.
- Developed custom jiggig systems to enhance precision for fabrication process.

Mouthguard Material Design | Queens University

Kingston, ON, March 2024 – April 2024

- Designed a custom mouthguard material using additive manufacturing in SOLIDWORKS to enhance athlete safety.
- Performed finite element analysis (FEA) on the design with 50N of force at the top of the design.
- 3D printed design completed the weighted drop test requirement, successfully enhancing safety properties.